Hospital Management System Project Documentation

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1. Introduction

- The system helps manage hospital activities like keeping patient records, scheduling appointments, recording treatments, managing medications, handling billing, and tracking inventory.

2. Project Overview

- The Hospital Management System is designed to make hospital processes easier, improve patient care, and increase efficiency. The system has features for managing patient information, scheduling appointments, recording treatments, handling medications, processing bills and payments, and generating reports.

3. Group Members

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4. Business Requirements

Functional Requirements:

1. Patient Management:
    - Create, update, delete, and view patient records.
    - Capture detailed patient information including personal details.
2. Appointment Management:

- Schedule, reschedule, and cancel patient appointments.
- Check for conflicts in doctor’s schedules and provide available time slots.

3. Treatment and Medication Management:
    - Record and update treatment details for each patient.
    - Manage medication inventory, including adding, updating, and tracking stock levels.
    - Generate prescriptions and link them to patient records.
4. Billing and Payments:
    - Generate bills based on treatments and medication provided.
    - Track payment status and history for each patient.
5. Inventory Management:
    - Track and manage the stock of medicines and supplies.
    - Update inventory levels automatically when medications are administered or dispensed.
6. Reporting:
    - Generate various reports including patient records, appointment schedules, billing details, and inventory status.

Non-Functional Requirements:

1. Performance:
    - The system should handle multiple concurrent users without significant performance degradation.
    - Data retrieval and processing should be quick and efficient.
2. Usability:
    - The user interface should be intuitive and easy to navigate.
    - Provide user training and support documentation.
3. Reliability:
    - Ensure high system availability with minimal downtime.
    - Implement regular data backups and recovery procedures.
4. Security:
    - Ensure data confidentiality and integrity by implementing access controls and encryption.
    - Regularly update the system to protect against security vulnerabilities.
5. Scalability:

- The system should be scalable to handle increasing amounts of data and users.
- Support for future enhancements and integrations with other systems.

6. Maintainability:
   - The system should be easy to maintain and update.
   - Provide clear documentation for system configuration and maintenance procedures.

5. Mockup Screens

New Patient Creating Screen

This screen allows hospital administrators to efficiently input and store new patient information, ensuring that all necessary data is captured for medical records and administrative purposes.

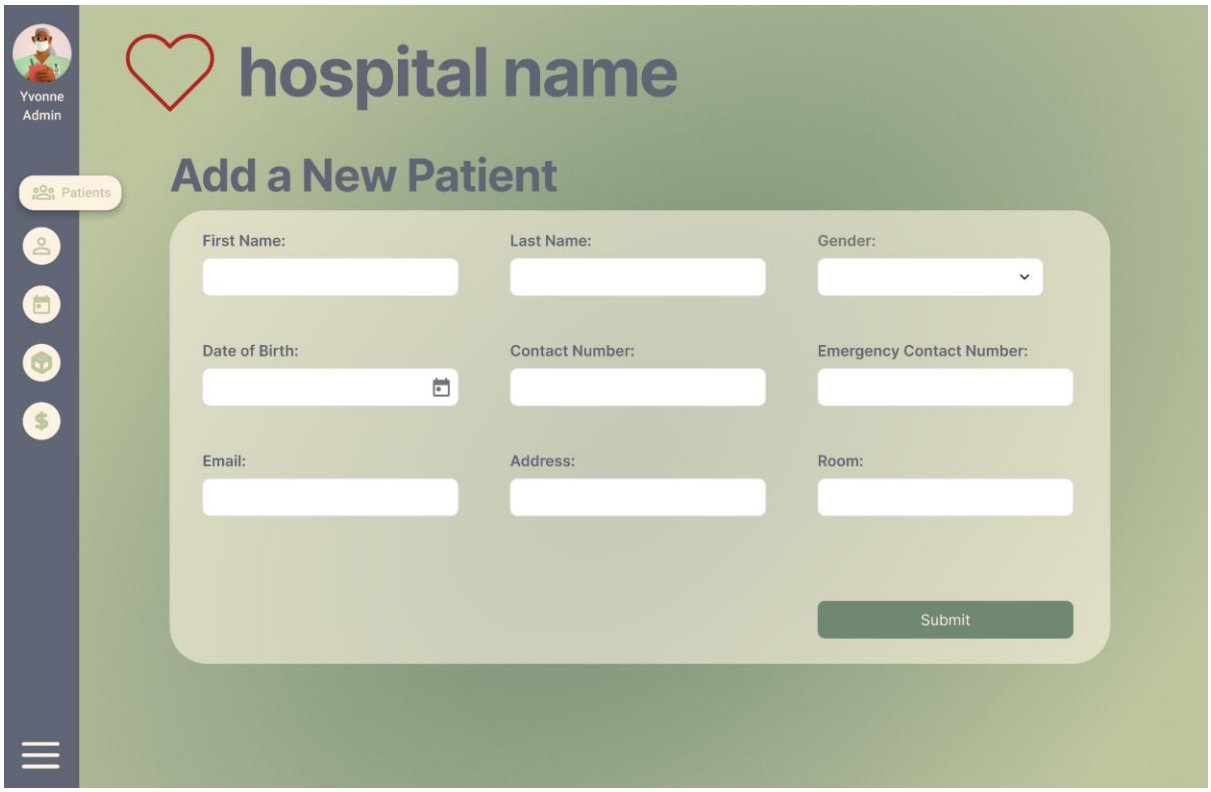


Figure 1-1: a mock-up screen of adding a new patient

If the data is all inserted correctly, a window will pop up and show the message.

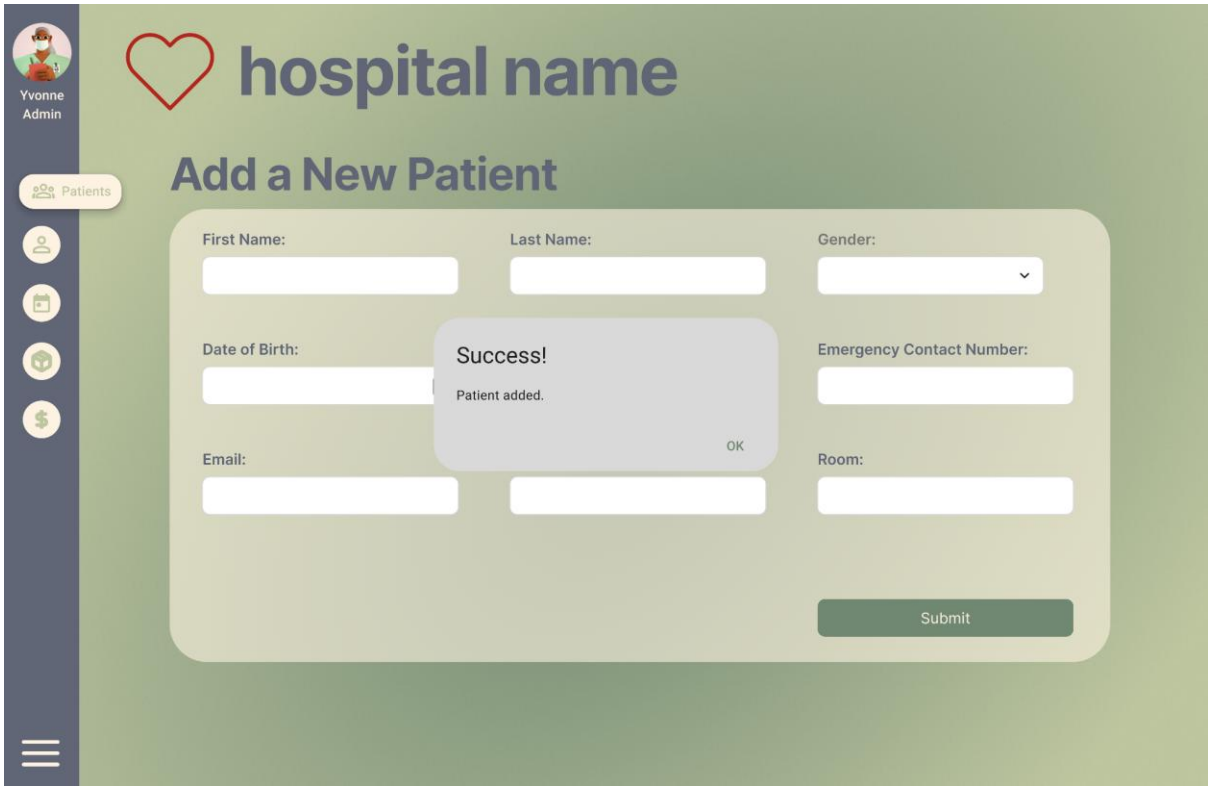


Figure 1-2: a mock-up screen of adding a new patient successfully

If the room is occupied, a window will pop up and show that you failed to add this new patient.

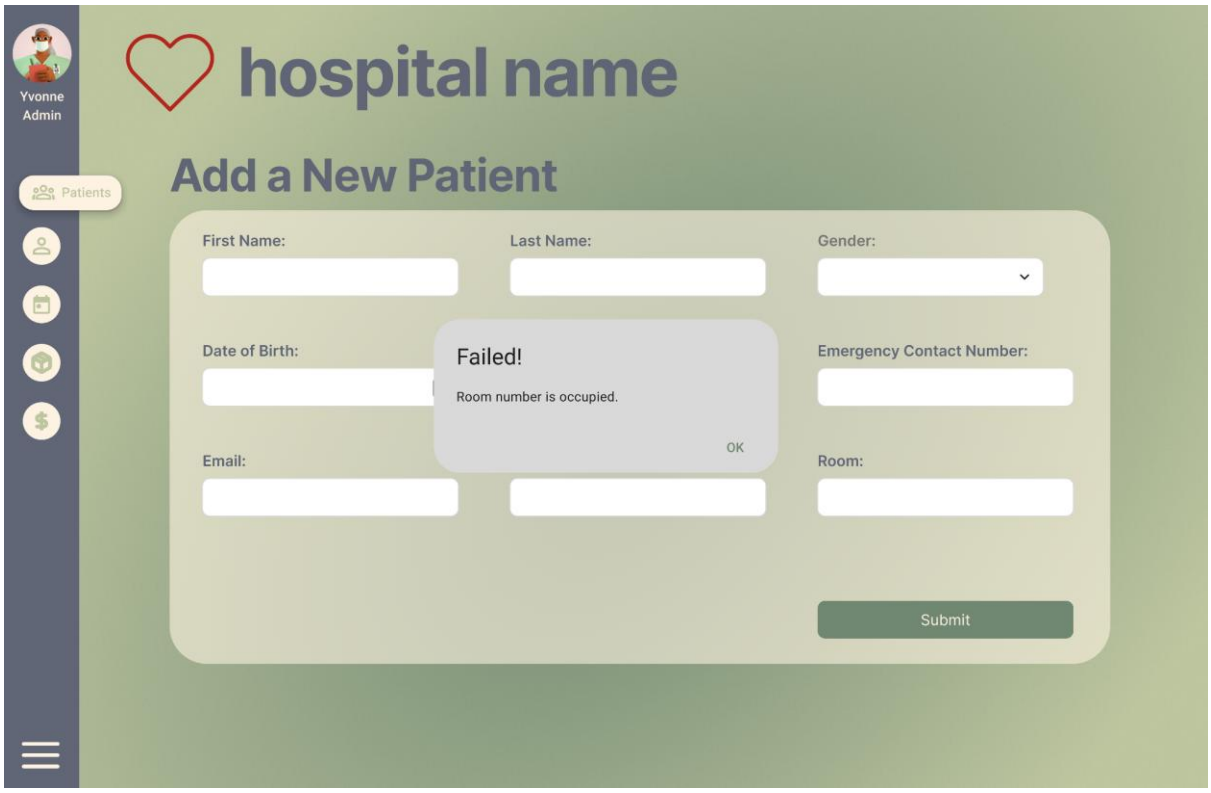


Figure 1-3: a mock-up screen of adding a new patient failed

Patient Searching Screen

The "Patient Searching Screen" facilitates searching and retrieving patient information within the hospital management system. It allows hospital administrators to locate and access patient records efficiently, enhancing workflow efficiency and patient care.

To avoid searching for another person with the same name, hospital administrators need to enter not only the patient's first and last name but also the patient's date of birth to ensure the search result is correct.

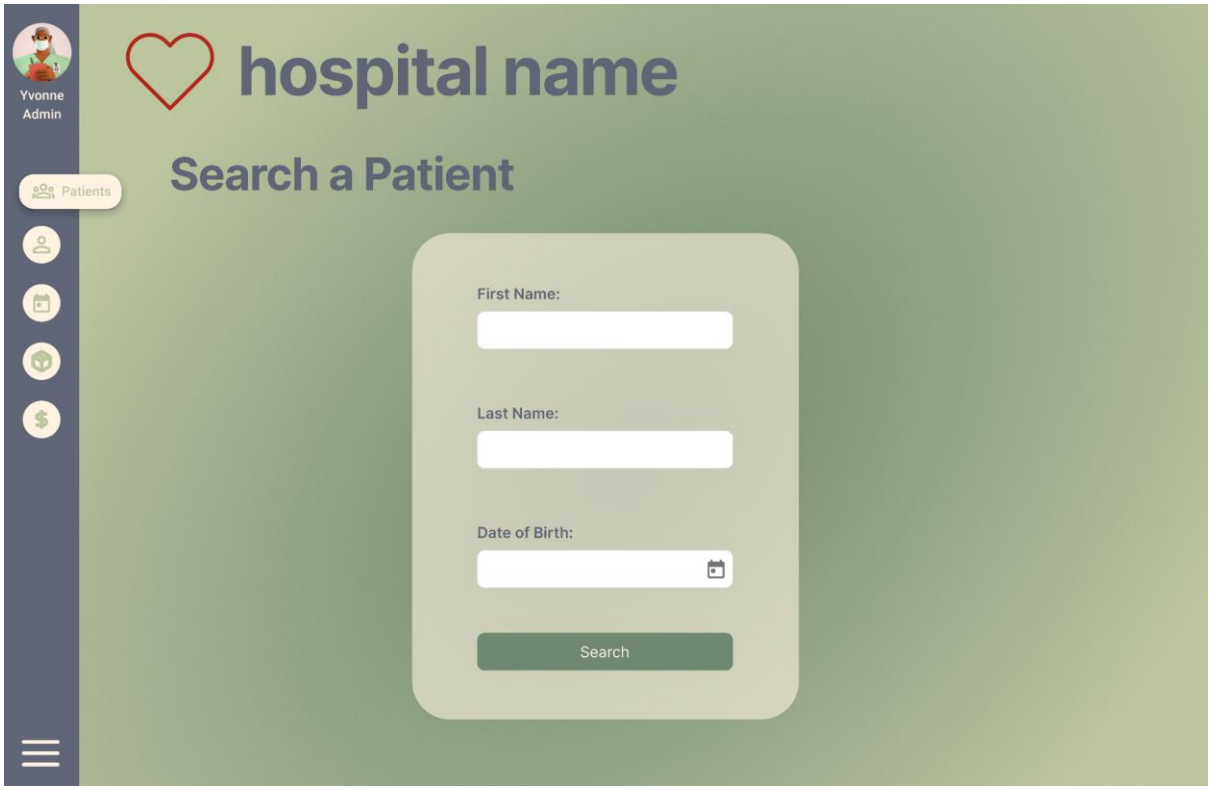


Figure 2-1: a mock-up screen of patient searching function

If the information matches the data in the hospital database, the system will show the patient profile.

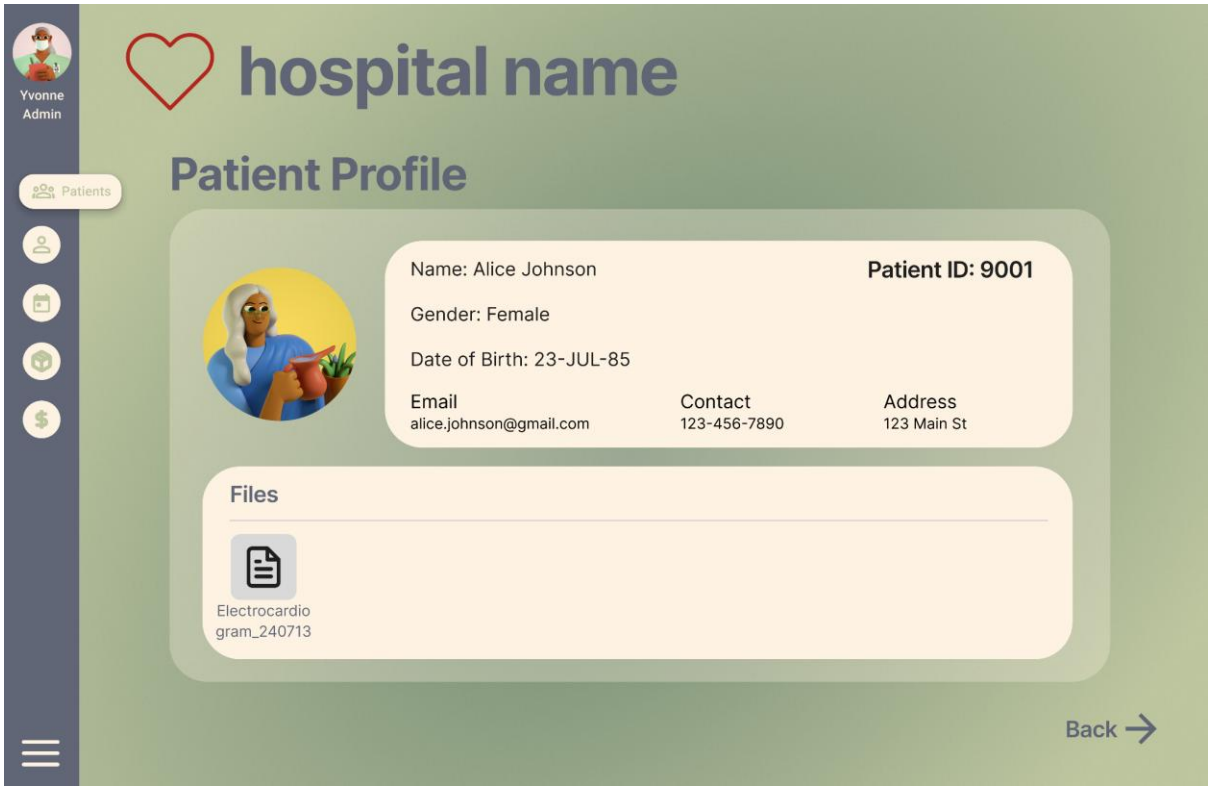


Figure 2-2: a mock-up screen of the patient profile

Appointment Scheduling Screen

The "Appointment Scheduling Screen" is designed to allow healthcare p roviders to schedule new appointments for patients. The screen ensures that no double booking occurs by checking for existing appointments at the same date and time for the selected doctor.

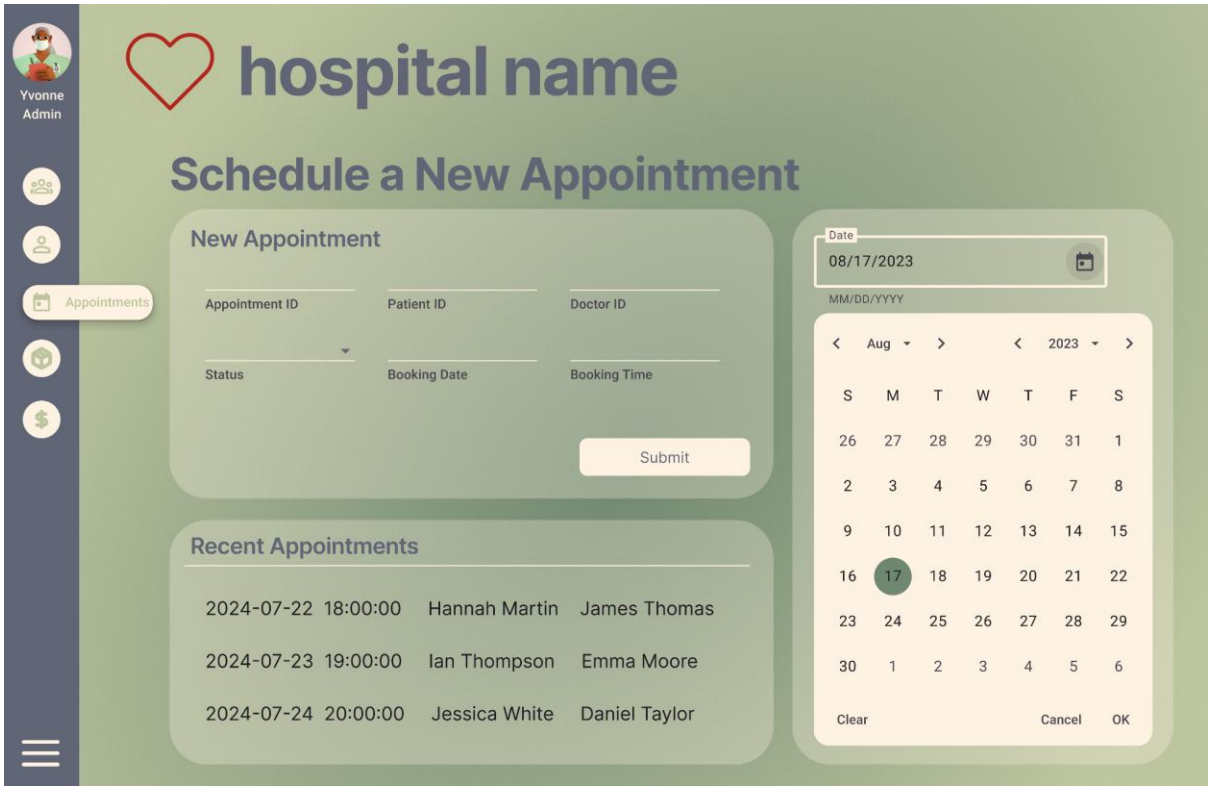


Figure 3-1: a mock-up screen of adding a new appointment

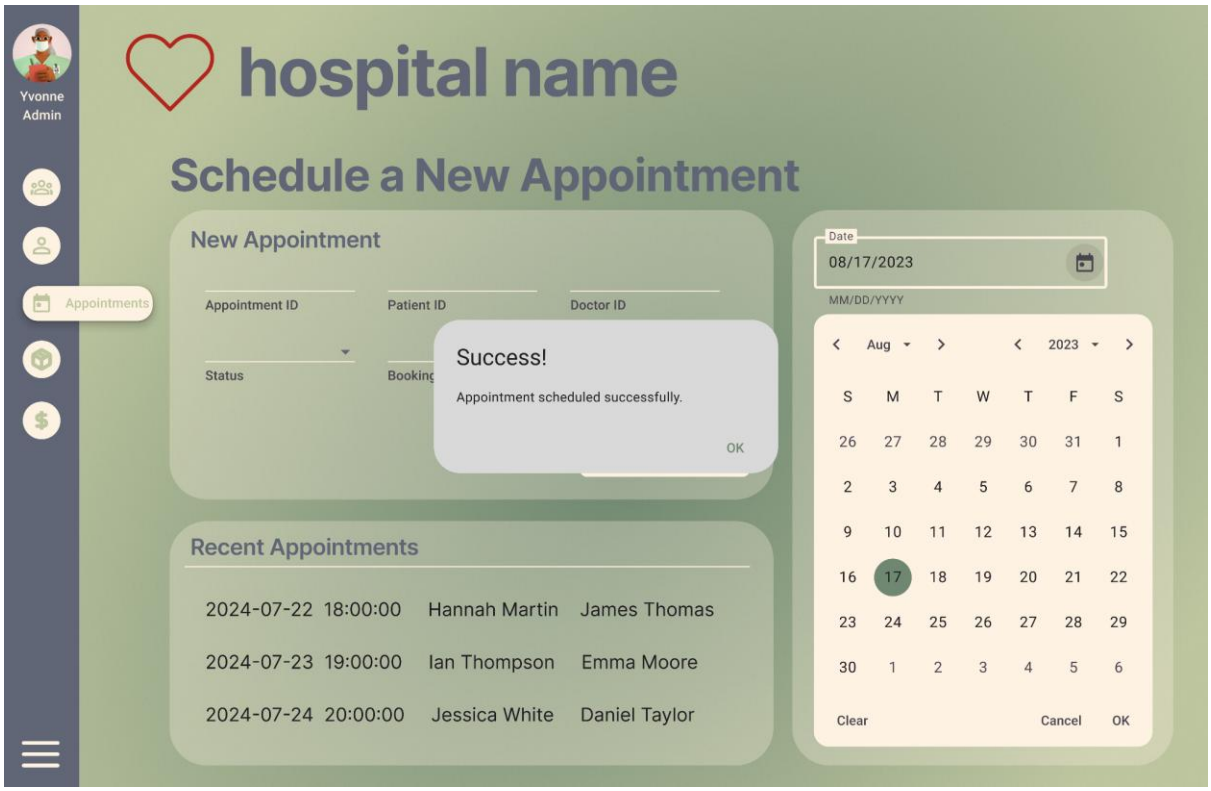


Figure 3-2: a mock-up screen of the appointment adding successfully

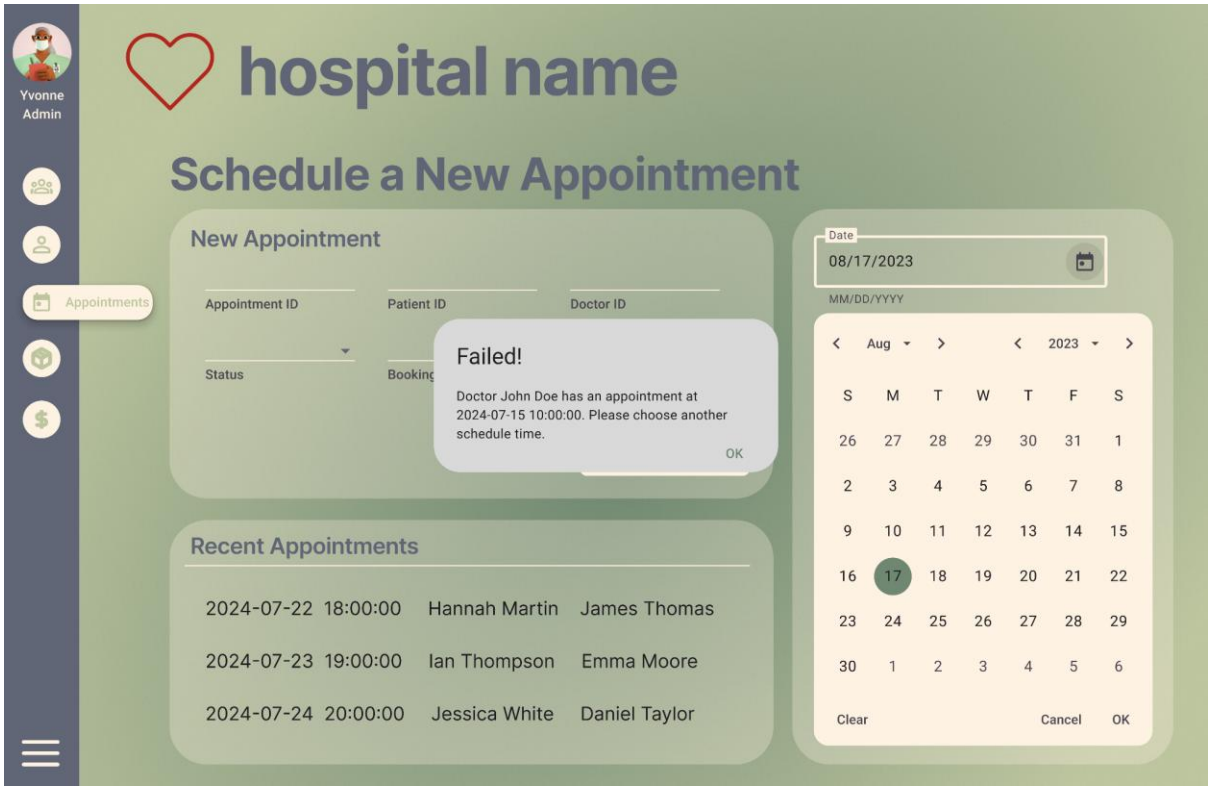


Figure 3-3: a mock-up screen of the appointment scheduling failed

Payments Checking Screen

The "Payments Checking Screen" allows users to enter a patient ID and view the total cost that the patient needs to pay, the total amount that the patient has paid, the amount that the patient still needs to pay, and the detailed payment history. This screen provides a comprehensive view of a patient's payment status, aiding in financial management and ensuring transparency in billing and payments.

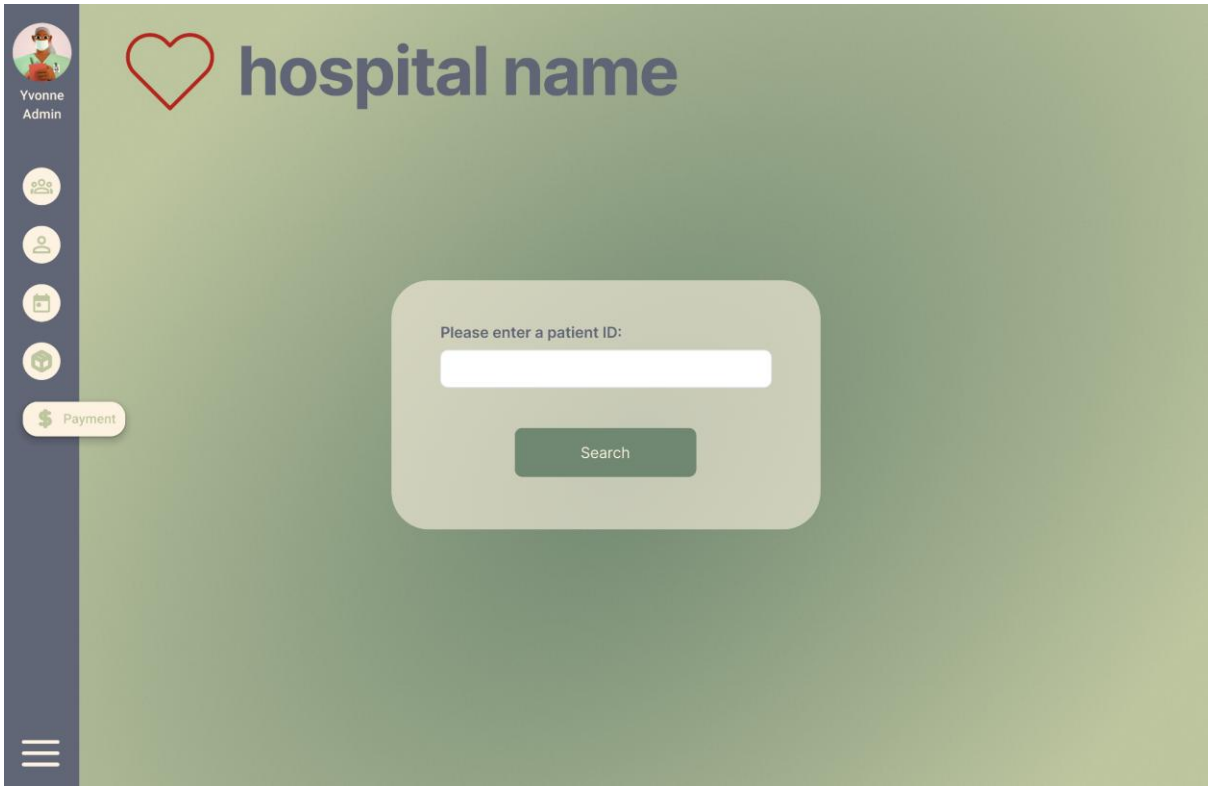


Figure 4-1: a mock-up screen of the payment checking

Figure 4-2: a mock-up screen of the payment history

Inventory Checking Screen (procedure)

The "Inventory Checking Screen" enables users to enter the name of a medicine and view its inventory details, including the name, stock quantity, expiration date, and manufacturer. This screen helps staff efficiently check and manage the inventory status of medicines, ensuring accurate stock levels and timely updates on medicine availability.

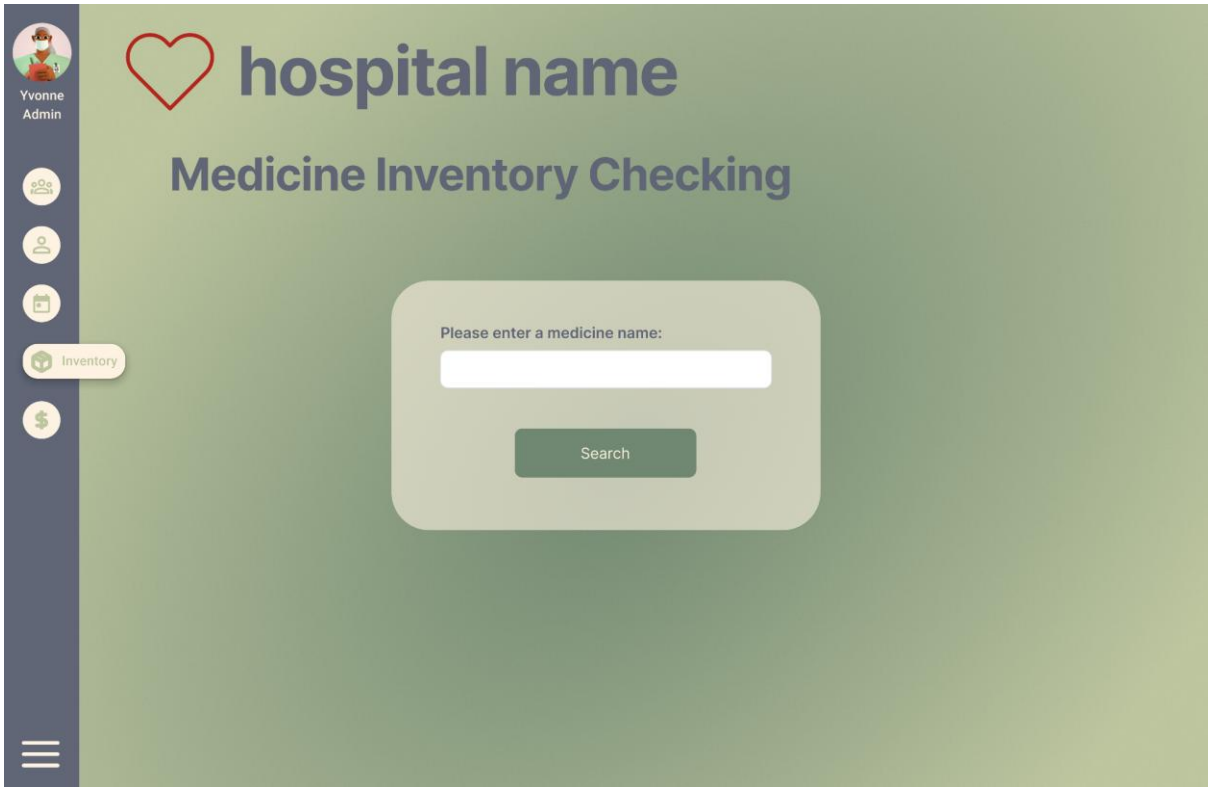


Figure 5-1: a mock-up screen of the inventory checking

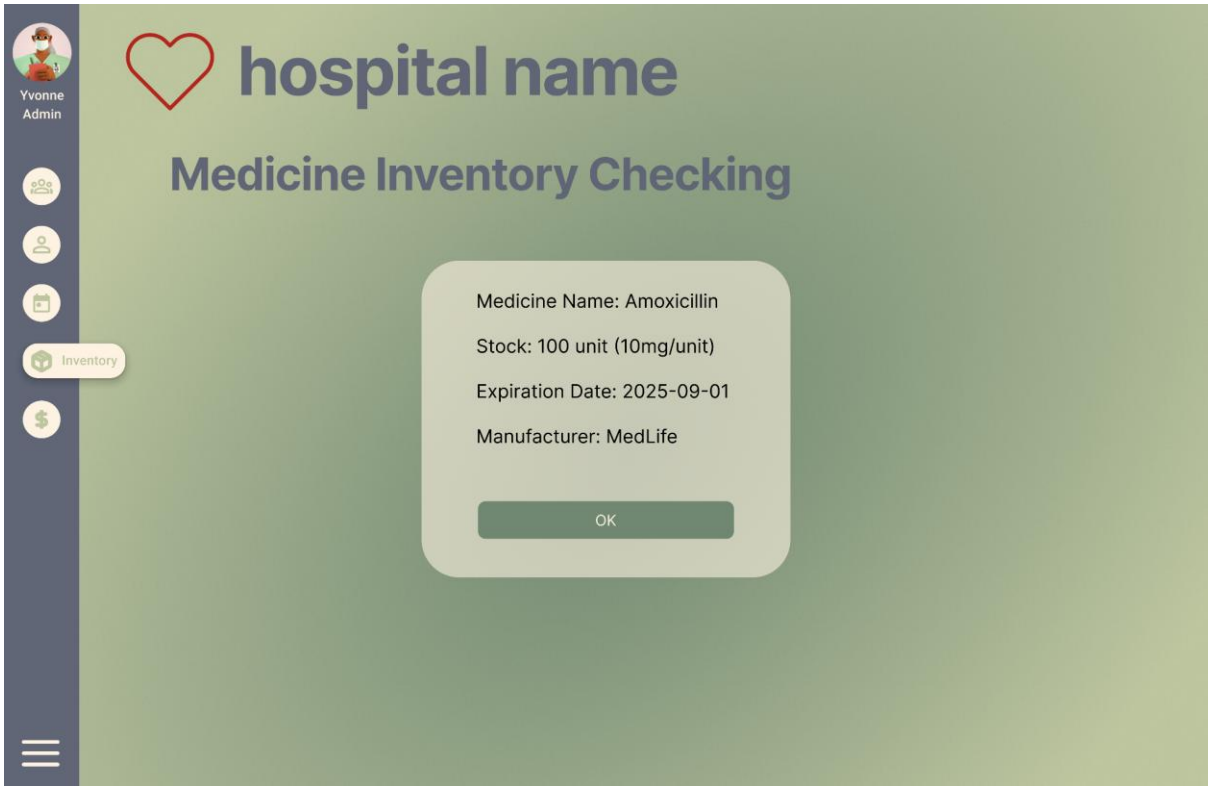


Figure 5-2: a mock-up screen of the inventory information

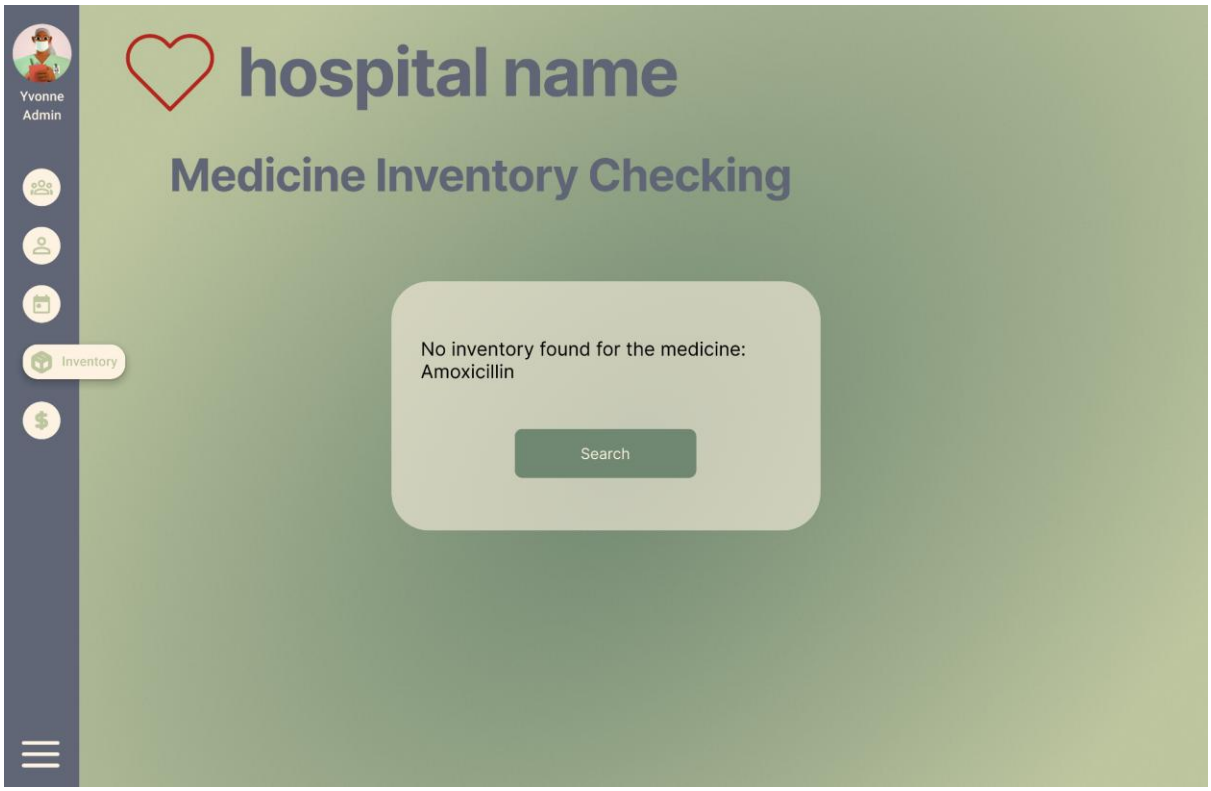


Figure 5-3: a mock-up screen of searching the medicine name failed

Inventory updating Screen (can use trigger)

The "Inventory Updating Screen" allows users to update the inventory details of medicines. When a new medication is administered or dispensed, the inventory is automatically adjusted to reflect the reduction in stock. This screen facilitates the real-time updating of the hospital's medicine inventory, ensuring accurate stock levels are maintained and any dispensed medications are accounted for immediately.

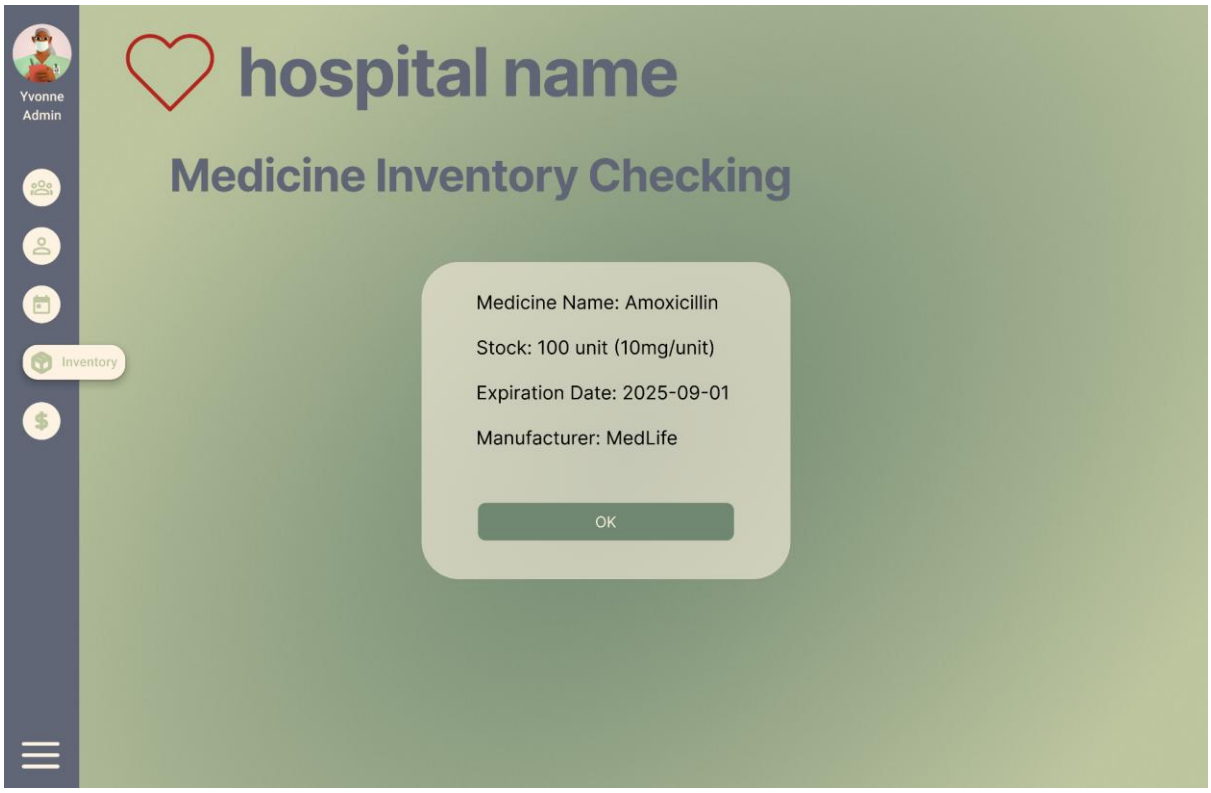


Figure 5-2: a mock-up screen of the inventory information

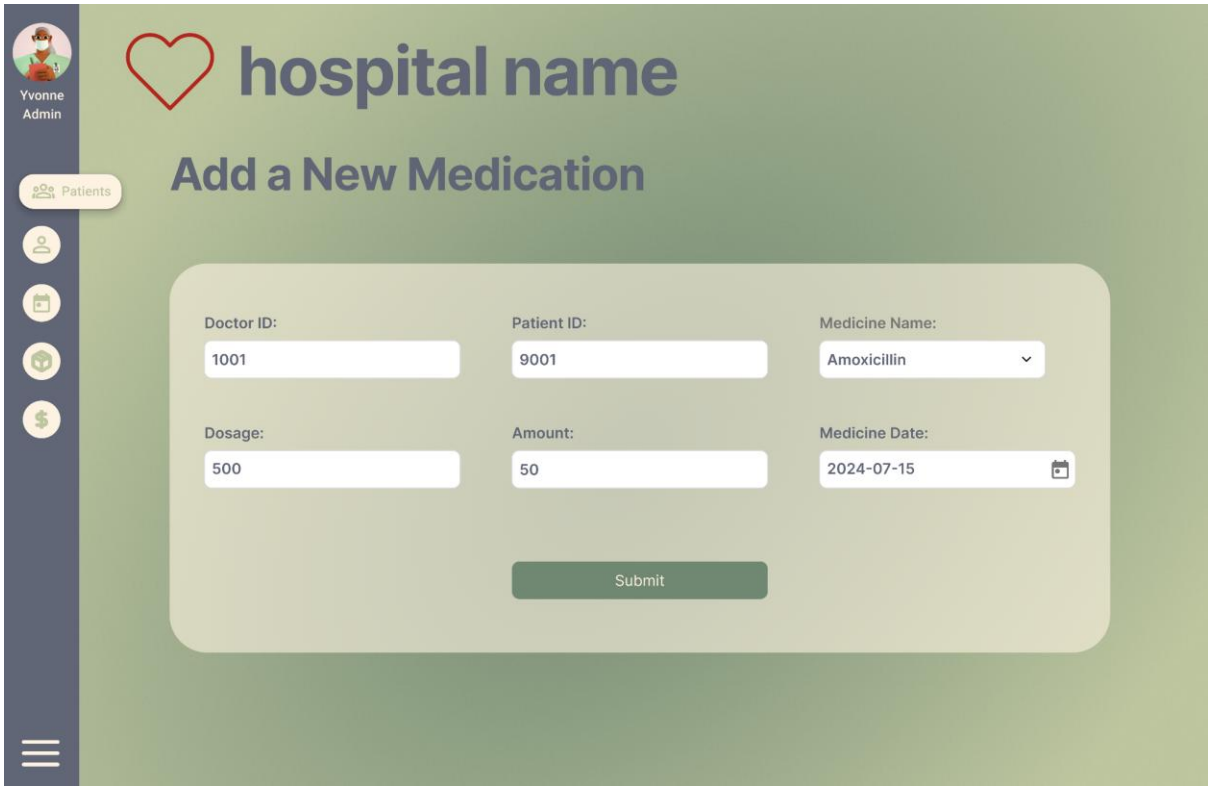


Figure 6-1: a mock-up screen of adding a new medication

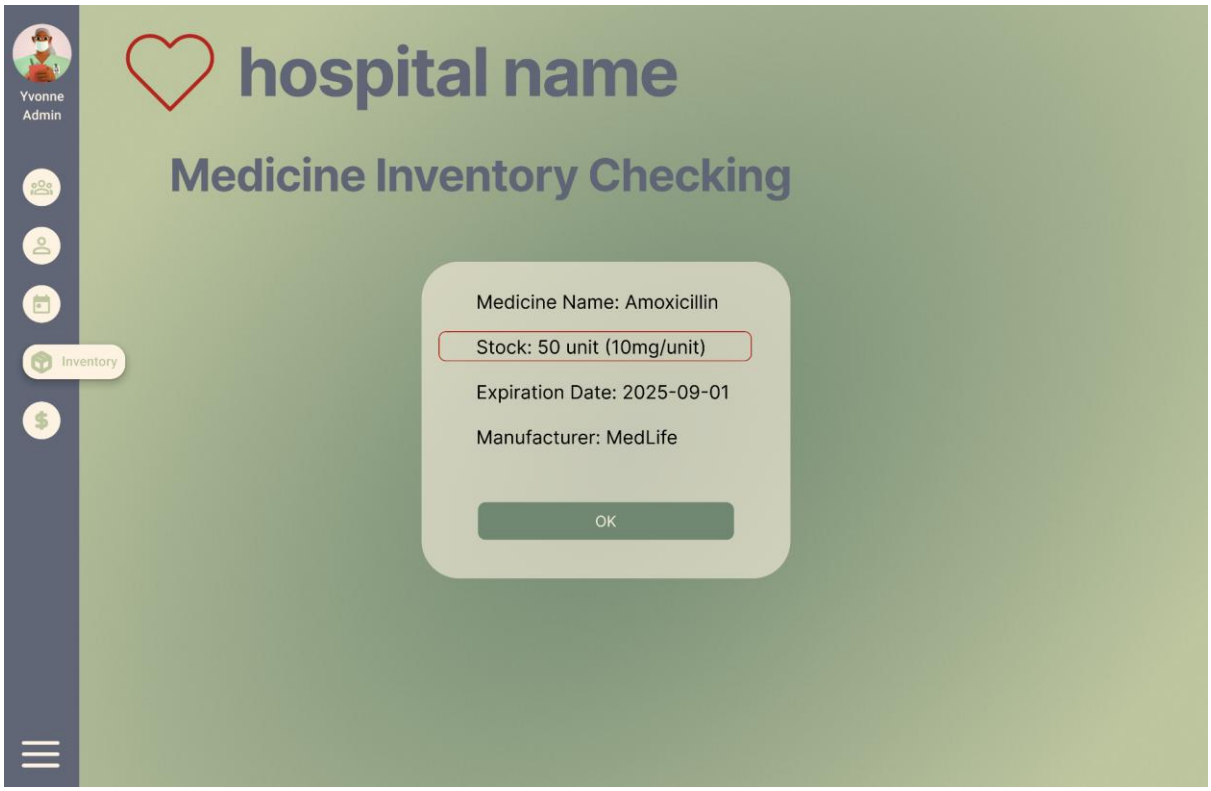
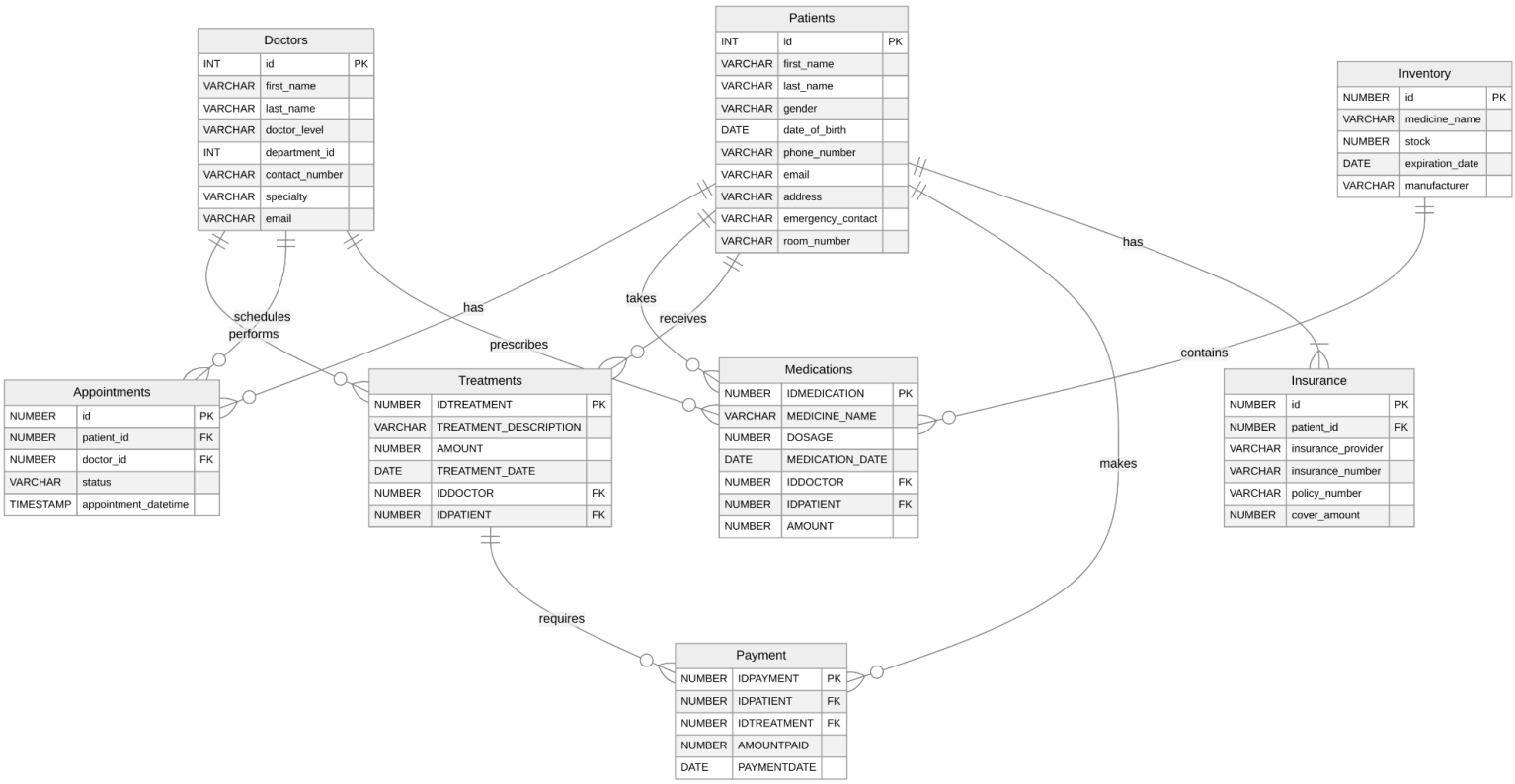


Figure 6-2: a mock-up screen of the inventory information after updating

6.ER Diagram

Include the database's Entity-Relationship Diagram (ERD), detailing the relationships between tables and key attributes.